

EAS 501 Lecture Notes 09/09/2026

Cheng Wang

Mathematics @ UMass Dartmouth

Outline

- Chapter 13. Differential Calculus and Functions of Several Variables

13.7 Maximum and minimum values

Theorem 13.7.1 Single-variable function case

If $f(x)$ has a local extreme at x_0 , and f is differentiable at x_0 , then $f'(x_0) = 0$.

Theorem 13.7.2 Higher order derivative test

If $f'(x_0) = f''(x_0) = \dots = f^{(n-1)}(x_0) = 0$, while $f^{(n)}(x_0) \neq 0$, $n \geq 2$. If n is even, we have the following classification

- ▶ f has a local maximum at x_0 if $f^{(n)}(x_0) < 0$;
- ▶ f has a local minimum at x_0 if $f^{(n)}(x_0) > 0$;

If n is odd, f has a horizontal inflection point.

Theorem 13.7.3 Multi-variable function case

If $f(x_1, x_2, \dots, x_n)$ has a local extreme at $\mathbf{x} = (x_0^{(1)}, x_0^{(2)}, \dots, x_0^{(n)})$, then $f_{x_1} = f_{x_2} \dots = f_{x_n} = 0$ at $\mathbf{x}$.

Theorem 13.7.4 For $f(x_1, x_2, \dots, x_n)$ with $f_{x_1} = f_{x_2} = \dots = f_{x_n} = 0$ at $\mathbf{x}$, we define the Hessian matrix

$$A = \begin{pmatrix} f_{x_1 x_1}(\mathbf{x}) & f_{x_1 x_2}(\mathbf{x}) & \dots & f_{x_1 x_n}(\mathbf{x}) \\ f_{x_2 x_1}(\mathbf{x}) & f_{x_2 x_2}(\mathbf{x}) & \dots & f_{x_2 x_n}(\mathbf{x}) \\ \dots & \dots & \dots & \dots \\ f_{x_n x_1}(\mathbf{x}) & f_{x_n x_2}(\mathbf{x}) & \dots & f_{x_n x_n}(\mathbf{x}) \end{pmatrix}.$$

Let $\det A \neq 0$.

- ▶ If A is positive definite, f has a local minimum at $\mathbf{x}$;
- ▶ If A is negative definite, f has a local maximum at $\mathbf{x}$;
- ▶ If A has at least one positive eigenvalue, one negative eigenvalue, then f has a saddle at $\mathbf{x}$.

Constrained extrema and Lagrange multiplier

We aim to optimize $f(x_1, x_2, \dots, x_n)$ subject to a constraint $g(x_1, x_2, \dots, x_n) = 0$.

In fact, this is equivalent to the optimization of $f^* := f - \lambda g$.

Lagrange multiplier method

$$(*) \quad \begin{cases} \nabla f - \lambda \nabla g = \mathbf{0}, \\ g(x_1, x_2, \dots, x_n) = 0, \end{cases}$$

Example 13.7 # 16 (b):

Find the point on $x + y = -5$ that is closest to $(1, -2)$.

Solution. We aim to minimize $f(x, y) = (x - 1)^2 + (y + 2)^2$, subject to a constraint $g(x, y) = x + y + 5 = 0$.

A detailed calculation implies that

$$f_x = 2(x - 1), \quad g_x = 1, \quad f_y = 2(y + 2), \quad g_y = 1.$$

In turn, system (*) becomes

$$2(x - 1) - \lambda \cdot 1 = 0, \quad \text{which gives } x = 1 + \frac{\lambda}{2}, \quad (1)$$

$$2(y + 2) - \lambda \cdot 1 = 0, \quad \text{which gives } y = -2 + \frac{\lambda}{2}, \quad (2)$$

$$x + y + 5 = 0. \quad (3)$$

Example 13.7 # 16 (b), continued

A substitution of (1) and (2) into (3) results in

$$1 + \frac{\lambda}{2} + \frac{\lambda}{2} - 2 + 5 = 0, \quad \lambda + 4 = 0, \quad \lambda = -4.$$

Going back (1) and (2), we get $x = 1 + \frac{\lambda}{2} = -1$,
 $y = -2 + \frac{\lambda}{2} = -4$.

Answer: The point $(-1, -4)$ is the point on $x + y = -5$ that is closest to $(1, -2)$.

13.8 Leibniz rules

Fundamental theorem of Calculus: $\frac{d}{dt} \int_a^t f(x) dx = f(t)$
Leibniz rule

$$\begin{aligned} \frac{d}{dt} \int_{a(t)}^{b(t)} f(x, t) dx &= \int_{a(t)}^{b(t)} f_t(x, t) dx \\ &\quad + b'(t)f(b(t), t) - a'(t)f(a(t), t). \quad (4) \end{aligned}$$

Example 13.8 # 1 (e)

Compute $\frac{d^2}{dx^2} \int_x^{2x} \ln(u^2 + x^2) du$

Solution. We set $f(u, x) = \ln(u^2 + x^2)$, so that $f_x = \frac{2x}{u^2 + x^2}$.

An application of Leibniz rule gives

$$\begin{aligned}
 & \frac{d}{dx} \int_x^{2x} \ln(u^2 + x^2) du \\
 = & \int_x^{2x} f_x(u, x) du + (2x)'_x f(2x, x) - (x)'_x f(x, x) \\
 = & \int_x^{2x} \frac{2x}{u^2 + x^2} du + 2 \cdot \ln(5x^2) - 1 \cdot \ln(2x^2) \\
 = & \left(2 \tan^{-1}\left(\frac{u}{x}\right) \right) \Big|_{u=x}^{u=2x} + 2(\ln 5 + 2 \ln x) - 1 \cdot (\ln 2 + 2 \ln x),
 \end{aligned}$$

Example 13.8 # 1 (e), continued

$$\begin{aligned}
 & \frac{d}{dx} \int_x^{2x} \ln(u^2 + x^2) du \\
 &= 2 \tan^{-1} 2 - 2 \tan^{-1} 1 + 2 \ln 5 - \ln 2 + 2 \ln x \\
 &= 2 \tan^{-1} 2 - \frac{\pi}{2} + \ln \frac{25}{2} + 2 \ln x.
 \end{aligned} \tag{5}$$

Then we arrive at

$$\frac{d^2}{dx^2} \int_x^{2x} \ln(u^2 + x^2) du = 0 + 2(\ln x)'_x = 2 \cdot \frac{1}{x} = \frac{2}{x}. \tag{6}$$